

SUPPLEMENTARY MATERIALS

Paper 1: Beyond resistance-gene calling: a provenance-aware case study of ciprofloxacin

genotype–phenotype discordance in African *Escherichia coli*

Publication-facing supplementary draft v3 | 7 September 2026

This reader-facing supplement was distilled from a provenance-locked S1-S16 evidence packet. No new biological analysis, threshold tuning, candidate discovery, association testing or causal inference was performed during editorial preparation.

The complete machine-readable TSVs, scripts, SHA256 manifests and terminal transcripts remain in the project provenance archive. Historical workflow states (for example, “Next gate” or “Paper1 complete = NO”) and absolute local computer paths are intentionally omitted from this publication-facing document because they are audit metadata rather than final scientific results.

Reporting authority: technical unavailability and technical unresolution are always kept distinct from biological negatives. The external *ompF* result is reported as 0/5 callable, with CIP06 technically unresolved. The external *qnrB* result is reported as 0/2 technically evaluable under frozen thresholds, with CIP02-CIP05 technically unavailable.

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S1. Dataset and isolate provenance

The focal case was ID005, a ciprofloxacin-resistant *E. coli* isolate from the Tanzania component of DSET004. ID003, a ciprofloxacin-susceptible isolate from the same source cohort, was retained as a close-background comparator; it was not treated as an exact clonal match. The external contextualisation panel comprised six ciprofloxacin-resistant isolates from Nigeria and Egypt and was used only for descriptive recurrence context.

Table S1A. Focal isolate and close-background comparator.

Alias	Role	Dataset/ country	CIP	BioSample	Assembly	Run	Comparator relationship
ID005	Focal target	DSET004 Tanzania	R	SAMN285906 61	GCA_0321855 45.1	SRR19349300	Target
ID003	Comparator	DSET004 Tanzania	S	SAMN285805 21	GCA_0321855 25.1	SRR19349003	Close- background; not exact clonal match

Table S1B. External CIP-R contextualisation panel and final system-specific evaluability.

ID	Country	Isolate	BioSample	Primary sequence record	ompF status	qnrB status
CIP01	Nigeria	OAU-AAA-018a	SAMEA6155699	ERR4783067	Callable	Technically evaluable
CIP02	Egypt	Me20	SAMN46163129	JBKQDE0000000 00	Callable	Technically unavailable
CIP03	Egypt	Me26	SAMN46163133	JBKQDI0000000 00	Callable	Technically unavailable
CIP04	Egypt	Me34A	SAMN46163140	JBKQDP0000000 00	Callable	Technically unavailable
CIP05	Egypt	Me35B	SAMN46163141	JBKQDQ0000000 000	Callable	Technically unavailable
CIP06	Egypt	Me82	SAMN46163168	JBKQER0000000 00; SRR39261344 resolved for raw reads	Technically unresolved	Technically evaluable

Note. The external panel contained only CIP-R isolates. It therefore supports descriptive recurrence contextualisation, not resistant-versus-susceptible association testing.

S2. Software, databases and reference resources

Only versions and database identities verified in the frozen project records are reported here. Complete executable paths and environment snapshots remain in the provenance archive.

Table S2. Software, database and reference register.

Resource	Version / identifier	Frozen detail	Use
AMRFinderPlus	4.2.7	Database 2026-08-07.1	Assembly-based AMR/QRDR screening; acquired-quinolone reference freeze
ResFinder	4.7.2	DB commit eecf0aa207594fe6d51badf80847 3de62b28cb06	Acquired resistance-gene screening
PointFinder	4.1.1	DB commit 44ce624a806c6d2b70f7e39841a 5f9cb4d9010aa	Known and unclassified gyr/par mutation screening
CARD/RGI	RGI 6.0.8 / CARD 4.0.2	Protein mode; Prodigal 2.6.3 protein prediction; frozen RGI metadata records BLAST alignment engine	Independent curated resistance-model comparison
FastQC	0.12.1	Frozen environment	Raw-read QC
BWA	0.7.19-r1273	Frozen environment	Own-reference and candidate-panel mapping
SAMtools	1.24	HTSlib 1.24	BAM processing, depth and QC
BCFtools	1.24	HTSlib 1.24	Targeted pileup/minor-allele workflow
Python	3.11.16	Frozen analysis environment	Pipeline orchestration and parsing
E. coli K-12 MG1655	NC_000913.3	Reference coordinates for regulatory/structural contexts	Coordinate/reference framework
qnrB focal reconstruction reference	qnrB1; NG_050469.1	645 bp; SHA256 ce10b5d7fbfdd640f6a6ec6bb740 bf6f874d640dc460f5dc9566503 5415d8ec6	Fixed reference for ID005 qnrB locus reconstruction

Note. The acquired-quinolone reference universe comprised 203 unique sequences across 12 included families.

The full sequence manifest is retained as a machine-readable supplementary TSV rather than reproduced as a 203-row print table.

S3. Known quinolone determinants and QRDR audit

Assembly-based screening did not retain a recognized acquired quinolone determinant or canonical QRDR mutation for ID005. PointFinder reported three unclassified substitutions (gyrA D678E, parC D475E and parC Q695L), but these were not recognized resistance-associated QRDR calls. CARD/RGI

reported MarR G103S/Y137H; subsequent comparator-aware coding analysis showed that the audited MarR protein was identical between ID005 and ID003, so these reference-relative MarR calls were not retained as ID005-specific candidates.

Table S3. Cross-tool quinolone/QRDR findings and final adjudication for ID005.

Evidence source	Finding	Final adjudication
AMRFinderPlus	No acquired quinolone determinant; no recognized QRDR determinant	No high-priority candidate retained from this branch
ResFinder	No acquired quinolone determinant	No high-priority candidate retained
PointFinder	No known gyr/par resistance mutation; unclassified gyrA D678E, parC D475E, parC Q695L	Unclassified substitutions retained as context only; not recognized QRDR resistance calls
CARD/RGI	MarR G103S/Y137H resistance-model match	Not ID005-specific after comparator-aware coding audit; not retained as a causal candidate

Note. The final adjudication is later than the initial cross-database status table and therefore supersedes its transient “unresolved” wording.

S4. Thirteen-gene coding audit

Thirteen prespecified regulator, porin and efflux proteins were assessed across the first-pass and second-tier coding audits. All 13 ID005 proteins were identical to the corresponding ID003 proteins, and no high-priority amino-acid coding candidate was retained.

Table S4. Final focal-pair coding endpoint across 13 audited genes.

Gene	Audit tier	ID005-ID003 result	Final disposition
marR	First-pass	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
acrR	First-pass	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
soxR	First-pass	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
ompF	First-pass	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
ompC	First-pass	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
marA	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
soxS	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
rob	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate

Gene	Audit tier	ID005-ID003 result	Final disposition
ompR	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
envZ	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
acrA	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
acrB	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate
tolC	Second-tier	Protein identical in ID005 and ID003	0 retained high-priority coding candidate

Note. Expression-dependent efflux/porin states were not resolved by sequence comparison and remain outside the causal claims of this study.

S5. Five-unit noncoding regulatory audit

Five prespecified physical regulatory units were compared. Four units had no target-specific physical event. NCUNIT05 (asnS-ompF context) retained four ID005-specific substitutions relative to ID003. One substitution overlapped the prespecified OmpR F1 active core; the other three were outside prespecified active ompF features. No functional effect was inferred.

Table S5. Retained ID005 ompF-region noncoding candidates.

Candidate	Local position	MG1655 position	Reference allele	ID005 allele	Feature context
NC_CAND001	201	987183	C	T	Overlaps OmpR F1 active core
NC_CAND002	273	987255	T	A	Outside prespecified ompF active cores
NC_CAND003	507	987489	C	T	Outside prespecified ompF active cores
NC_CAND004	580	987562	C	T	Outside prespecified ompF active cores

Note. These are sequence-location candidates only. Their external recurrence is reported separately in S12.

S6. Structural and insertion-sequence audit

Eight matched empirical proximal structural-context units were evaluated prospectively for pairwise events of at least 50 bp. All eight direct comparisons resolved. Six were exactly identical; two were non-identical but subthreshold (SCU01, 0-bp length difference; SCU06, +2 bp in ID005). No internal pairwise gap of at least 50 bp and no retained structural candidate was observed.

Table S6. Final structural/IS branch summary.

Metric	Final value
Matched structural-context units	8
Resolved direct alignments	8/8
Exact sequence-identity units	6
Non-identical but sub-50-bp units	2 (SCU01 and SCU06)
Internal ≥ 50 -bp pairwise gap observations	0
Resolved ≥ 50 -bp local structural events	0
Insertion-sequence family classification triggered	No
Residual structural candidates	0

Note. The negative endpoint applies only to the prospectively frozen ≥ 50 -bp local assembly-addressable structural event class. It does not prove that all smaller or assembly-inaccessible structural variation is absent.

S7. Raw-read quality control and mapping

Paired reads for ID005 and ID003 passed integrity checks and were mapped to each isolate's own frozen assembly using BWA-MEM. Duplicates were marked but retained in the common BAM; the primary copy-number depth profile excluded duplicate, secondary, supplementary, QC-failed and unmapped reads.

Table S7A. Raw-read QC summary.

Isolate	Run	Read pairs	Read length	R1 Q30 (%)	R2 Q30 (%)
ID005	SRR19349300	2,691,567	250 bp	91	93
ID003	SRR19349003	2,912,779	250 bp	94	93

Table S7B. Own-reference mapping/depth QC summary.

Isolate	Primary mapped fraction	Properly paired fraction	Duplicate fraction	Primary median depth	Breadth $\geq 10x$	Technical status
ID005	0.972098	0.881003	0.206841	144x	0.974322	Accepted / CN eligible
ID003	0.990957	0.916340	0.234498	182x	0.970670	Accepted / CN eligible

Note. Mapping acceptance required a primary mapped fraction ≥ 0.90 . Copy-number technical eligibility required primary nonduplicate MQ30 median depth $\geq 30x$ and ACGT breadth at $10x \geq 0.90$.

S8. Thirteen-gene copy-number screen

Copy-number screening used within-isolate local depth normalisation and a comparator-aware four-criterion rule. All four candidate criteria were frozen at 1.50 before the relevant depth values were interpreted. No gene passed all four criteria.

Table S8. Comparator-aware primary copy-number screen.

Gene	ID005 local norm.	ID003 local norm.	ID005/ID003 ratio	All four 1.50 criteria	Final classification
marR	0.967	1.098	0.881	Fail	No target-specific depth CN elevation
acrR	0.943	0.966	0.977	Fail	No target-specific depth CN elevation
soxR	1.133	1.060	1.068	Fail	No target-specific depth CN elevation
ompF	0.903	0.988	0.914	Fail	No target-specific depth CN elevation
ompC	0.972	0.949	1.024	Fail	No target-specific depth CN elevation
marA	0.884	0.948	0.933	Fail	No target-specific depth CN elevation
soxS	0.933	1.047	0.892	Fail	No target-specific depth CN elevation
rob	1.080	1.164	0.928	Fail	No target-specific depth CN elevation
ompR	1.153	1.126	1.025	Fail	No target-specific depth CN elevation
envZ	0.901	0.990	0.910	Fail	No target-specific depth CN elevation
acrA	1.092	1.232	0.886	Fail	No target-specific depth CN elevation
acrB	1.077	1.056	1.020	Fail	No target-specific depth CN elevation
tolC	1.000	1.035	0.966	Fail	No target-specific depth CN elevation

Note. A negative screen means no target-specific raw-read depth copy-number elevation under the frozen assay. It is not proof that every form of copy-number variation is absent, and no resolved amplification was inferred.

S9. Forty-two-object minor-allele screen

The targeted minor-allele universe comprised 42 isolate-specific objects spanning *gyrA*, *gyrB*, *parC*, *parE*, the 13 audited regulator/porin/efflux genes, and the four frozen *ompF* noncoding positions. Technical eligibility required MAPQ ≥ 30 , base quality ≥ 30 and filtered depth ≥ 100 . A preliminary

sequence-level minor-allele signal additionally required at least 10 alternate reads, VAF 0.10-0.40, and at least three alternate reads on each strand.

Table S9. Final targeted minor-allele screen.

Metric	ID005	ID003 / combined
Frozen target objects	21	21 (42 total)
Raw sites classified	22,389	22,389 (44,778 total)
Concrete physical allele events	4,356	3,067
Preliminary passing events	0	0
Target-specific signals declared	0	Not applicable
Branch endpoint	Negative under frozen thresholds	Negative under frozen thresholds

Note. This negative targeted WGS screen does not prove absence of variation below the detection thresholds, minor populations outside the frozen target universe, or phenotypic heteroresistance.

S10. Acquired quinolone raw-read audit

Raw-read interrogation of acquired quinolone determinants began from a prospectively frozen 203-sequence/12-family panel. Three preliminary signals were adjudicated: ID005 AAC6Ib-family evidence, ID005 qnrB, and ID003 qnrS. Only ID005 qnrB remained as a target-side residual candidate at bounded closure.

Table S10A. Final disposition of acquired-quinolone raw-read signals.

Signal	Key evidence	Final disposition
ID005 AAC6Ib / aac(6')-Ib-cr	CR-resolved ends 1; non-CR-resolved ends 10; ambiguous ends 20; 0 technical unresolved; 0 CR-informative fragments; 9 non-CR-informative; 1 class-conflict fragment	Mixed/conflicting variant-class evidence; no single variant supported. Specific aac(6')-Ib-cr not supported; broad AAC6Ib family signal not retained as a target quinolone candidate.
ID005 qnrB	11 fragments; 20 supporting ends; 12 distinct starts; coherent 645-bp reference-guided locus	Retained residual sequence candidate; assembly-tool uncalled; genomic linkage unresolved
ID003 qnrS	Comparator signal represented by same-family assembly evidence	Comparator context only; not an ID005 target candidate

Table S10B. ID005 qnrB locus reconstruction and pair-context endpoint.

Metric	Value
Fixed reference	qnrB1 (NG_050469.1), 645 bp
Locked fragments / supporting ends	11 / 20
R1 / R2 supporting ends	11 / 9

Metric	Value
Minimum qualifying alignment	≥ 100 alignment columns; identity ≥ 0.95
Distinct reference starts	12
Breadth at depth ≥ 1	1.000000
Breadth at depth ≥ 2	1.000000
Breadth at depth ≥ 3	0.998450
High-confidence consensus breadth	0.998450 (depth ≥ 3 ; dominant AF ≥ 0.80 ; base quality ≥ 30)
High-depth mixed positions	0
Indel observations	0
Context mates evaluated	2
Unique assembly-context anchors	0
Technically unresolved context mates	2
Final linkage status	Unresolved
Publication wording for genomic presence	Genomic qnrB presence was not established because genomic linkage remained unresolved

Note. The qnrB reconstruction is strong raw-read locus evidence, but it does not establish chromosomal/plasmid location, insertion site, mobility, transferability, function or ciprofloxacin causality.

S11. External contextualisation cohort and input evaluability

The external contextualisation cohort was frozen as six CIP-R isolates before result generation. All six had frozen assembly inputs for ompF contextualisation. For qnrB, public raw-read accession resolution ultimately produced a technically evaluable denominator of CIP01 and CIP06; CIP02-CIP05 remained technically unavailable.

Table S11. Final external system-specific evaluability.

ID	CIP	Assembly for ompF	Final ompF evaluability	Raw reads for qnrB	Final qnrB evaluability
CIP01	R	Frozen local SPAdes assembly	Callable	ERR4783067 paired reads	Technically evaluable
CIP02	R	GCA_050489365.1	Callable	No provenance-linked paired reads	Technically unavailable
CIP03	R	GCA_050489285.1	Callable	No provenance-linked paired reads	Technically unavailable
CIP04	R	GCA_050489145.1	Callable	No provenance-linked paired reads	Technically unavailable
CIP05	R	GCA_050489115.1	Callable	No provenance-linked paired reads	Technically unavailable
CIP06	R	GCA_050488505.1	Technically unresolved at ompF locus	SRR39261344 paired reads	Technically evaluable

Note. Earlier pre-acquisition snapshots in the audit trail are not reproduced here because they do not represent the final evaluability state.

S12. External ompF contextualisation

The fixed 601-bp MG1655 NCUNIT05 query and four prespecified ID005 positions were evaluated in the six external CIP-R assemblies. Five isolates resolved uniquely; CIP06 had no eligible locus and remained technically unresolved. None of the four exact ID005 alleles recurred among the five callable isolates.

Table S12A. Per-isolate ompF four-site system status.

ID	Locus status	ID005-allele count	System status
CIP01	Resolved unique highest-AS locus	0	No ID005-site recurrence
CIP02	Resolved unique highest-AS locus	0	No ID005-site recurrence
CIP03	Resolved unique highest-AS locus	0	No ID005-site recurrence
CIP04	Resolved unique highest-AS locus	0	No ID005-site recurrence
CIP05	Resolved unique highest-AS locus	0	No ID005-site recurrence
CIP06	Technically unresolved: no eligible locus	NA	Technically unresolved

Table S12B. Site-level external recurrence of the four ID005 ompF alleles.

Candidate	Callable n	Exact ID005 allele n	MG1655 reference n	Other allele n	Technical missingness	Interpretation
NC_CAND001	5	0	3	2 (A in CIP02, CIP03)	1 (CIP06)	Position polymorphic; exact ID005 T not observed
NC_CAND002	5	0	5	0	1 (CIP06)	Exact ID005 A not observed
NC_CAND003	5	0	5	0	1 (CIP06)	Exact ID005 T not observed
NC_CAND004	5	0	5	0	1 (CIP06)	Exact ID005 T not observed

Note. Required reporting: 0/5 callable external isolates for exact ID005 allele recurrence; CIP06 is technically unresolved and is not a biological negative.

S13. External qnrB contextualisation

The final qnrB denominator comprised only the two external isolates with provenance-resolved paired raw reads (CIP01 and CIP06). Neither produced a qnrB raw-read signal under the frozen thresholds. CIP02-CIP05 remained technically unavailable and were excluded from the negative denominator.

Table S13. Final external qnrB context status.

ID	Technical status	Resolved qnrB fragments	R1	R2	Breadth	Distinct starts	Final qnrB status
CIP01	Technically complete	0	0	0	0	0	Raw-read signal not detected under frozen thresholds
CIP06	Technically complete	0	0	0	0	0	Raw-read signal not detected under frozen thresholds
CIP02	Technically unavailable	NA	NA	NA	NA	NA	Not evaluated
CIP03	Technically unavailable	NA	NA	NA	NA	NA	Not evaluated
CIP04	Technically unavailable	NA	NA	NA	NA	NA	Not evaluated
CIP05	Technically unavailable	NA	NA	NA	NA	NA	Not evaluated

Note. Required reporting: 0/2 technically evaluable external isolates under frozen thresholds; CIP02-CIP05 technically unavailable. This must not be simplified to 0/6.

S14. Claims, denominators and limitations

This section states the reporting constraints that govern both the main manuscript and supplement. It is intentionally explicit because the scientific contribution of the case study depends on preserving uncertainty and system-specific technical missingness rather than forcing binary genotype assignments.

Table S14A. Permitted claims and prohibited overstatements.

Topic	Permitted publication claim	Do not overstate as
ID005 stopping state	Five sequence-level candidate items across two systems (ompF noncoding and qnrB) remained at bounded closure.	Do not describe these as five established resistance mechanisms.
ompF target-side evidence	Four ID005-specific ompF-region substitutions were retained; one overlaps the prespecified OmpR F1 active core.	Do not state that these substitutions caused ciprofloxacin resistance.

Topic	Permitted publication claim	Do not overstate as
ompF external context	None of the four exact ID005 alleles recurred among five callable external CIP-R isolates.	Do not report absence in all six; CIP06 was unresolved.
qnrB target-side evidence	A coherent assembly-tool-uncalled raw-read qnrB locus was retained with unresolved genomic linkage.	Do not state that a genomic qnrB gene was definitively established in ID005.
qnrB external context	No qnrB signal was detected in either technically evaluable external isolate.	Do not report qnrB absence in all six; CIP02-CIP05 were unavailable.
Association	External results are descriptive recurrence context among CIP-R isolates.	No R-vs-S association, odds ratio, sensitivity, specificity or predictive accuracy.
Causality	The causal basis of ID005 ciprofloxacin resistance remains unresolved.	No causal ranking of ompF versus qnrB and no functional/expression/plasmid inference.

Table S14B. Denominator and missingness registry.

Analysis	Nominal panel	Technically evaluable	Positive	Negative under frozen assay	Technical missingness	Required reporting
ompF exact ID005 allele recurrence	6	5	0	5	CIP06	0/5 callable; CIP06 technically unresolved
ompF full four-site haplotype	6	5	0	5	CIP06	0/5 resolved; CIP06 technically unresolved
qnrB raw-read signal	6	2	0	2	CIP02-CIP05	0/2 technically evaluable; CIP02-CIP05 unavailable

Table S14C. Principal limitations.

Limitation	Consequence
External Pilot6 panel contained only CIP-R isolates.	Cannot estimate resistant-versus-susceptible association or specificity.
CIP06 ompF locus was technically unresolved.	ompF denominator is five callable isolates, not six biological negatives.
CIP02-CIP05 lacked evaluable paired raw reads for qnrB.	qnrB denominator is two technically evaluable isolates.
ID005 qnrB genomic linkage remained unresolved.	Cannot assign genomic location, plasmid context, insertion site, mobility or definitive genomic presence.
Sequence-only design cannot resolve expression-dependent effects.	Efflux/porin regulation requiring expression evidence remains unresolved.
Phenotype uncertainty was retained.	Residual genotype-phenotype discordance must be interpreted with AST/data-quality uncertainty.
Pilot cohort was small and descriptive.	No formal association, predictive-performance or causal inference is supported.

S15. Reproducibility and provenance ledger

The full terminal transcripts and machine-specific paths remain in the audit archive. The publication supplement records only the gate identity, scientific role, final outcome and SHA256 so that the analysis can be traced without reproducing raw terminal noise.

Table S15A. Key successful analysis/reporting gates.

Gate	Role	Outcome	SHA256
ID005 sequence-only stopping state	Biological stopping rule	5 retained items / 2 systems; discovery closed	b2b4312a02f81f460274526e6f954cf620786db1f4193e938fbf72de91253650
External candidate integration	Context integration	Both external branches integrated without causal ranking	bcd06e0bf17afba510a129fa019a1e16fced40d3e51ed3def9dcb8f8f831aef
Paper1 reporting freeze	Reporting boundary	System-specific denominators and claim ceilings frozen	527fe24ea8c9d02b7422ab45ea16fa0301303d1df01b38886dca89a2ff2d17b0
Manuscript evidence pack	Manuscript evidence materialisation	Evidence pack complete	223bb22914b9d0f25d74adf4b3dd7d5b66249639abe065dfb785cf24da2fb4
Rendered manuscript outputs	Tables/figures materialisation	4 tables + 2 vector figures reproduced	b7dd75254202ab97c19f5ef1eb99f16c68ca242f36f2537e3be646711f78cc18

Table S15B. Preserved failed-history gates (provenance only).

Failed gate	Reason / disposition	SHA256
Pilot6 cohort-ontology gate	Failed metadata-resolution gate; superseded by repaired cohort architecture freeze	c556357560479d7ede3d84199d128a0f2ba1a531afd8432dc814180dc6c9a445
MG1655/context implementation gate	Failed implementation/provenance gate; retained only as audit history	6522345813f4a6a51e6bd72026d0b98ad9ce6233b6c9cb87a9e098d39f49bad7
CIP06 acquisition NameError gate	Pre-evidence acquisition failure; later repaired; no biological negative inferred	c1782d38af10c1b89b065dd1613a8064564bf41a76d68c84bdba18ff3f4a89a6

Note. Failed-history artifacts are not evidence for the biological endpoint. Their purpose is to preserve an auditable record of unsuccessful pre-evidence implementation states.

S16. Extended Methods and frozen decision rules

S16.1 Assembly-based AMR and QRDR screening

ID005 assembly-based quinolone resistance screening was cross-checked across AMRFinderPlus, ResFinder, PointFinder and CARD/RGI. Known acquired quinolone determinants and recognized QRDR mutations were treated separately from unclassified substitutions and reference-relative

resistance-model matches. Later comparator-aware coding analysis governed final candidate adjudication; therefore a reference-relative match was not retained if the corresponding ID005 protein was identical to ID003.

S16.2 Comparator-aware coding and noncoding audit

The coding audit comprised 13 prespecified regulator, porin and efflux genes: *marR*, *acrR*, *soxR*, *ompF*, *ompC*, *marA*, *soxS*, *rob*, *ompR*, *envZ*, *acrA*, *acrB* and *tolC*. Protein-level comparisons were interpreted against available susceptible or lineage-background comparators. For the focal pair, all 13 proteins were identical between ID005 and ID003. Noncoding interrogation used five prespecified physical units and retained only target-specific sequence differences after placement/adjudication. Functional consequences were not inferred from sequence location alone.

S16.3 Structural-context screen

Eight frozen matched empirical proximal contexts were compared for local assembly-addressable structural differences. The structural event class was prospectively defined as pairwise events ≥ 50 bp. Closure required all eight direct alignments to resolve and no internal ≥ 50 -bp pairwise gap. Sub-50-bp differences were recorded but not promoted into the structural candidate class. No insertion-sequence library was triggered because no resolved ID005-specific inserted sequence survived the ≥ 50 -bp gate.

S16.4 Raw-read QC and own-reference mapping

Frozen untrimmed paired reads were used after QC because no prespecified trimming trigger was activated. Each isolate was mapped to its own frozen canonical assembly with BWA-MEM. Duplicates were marked, not removed. Mapping acceptance required BAM integrity/primary-record consistency and a primary mapped fraction ≥ 0.90 . For copy-number technical eligibility, primary nonduplicate MQ30 ACGT median depth had to be $\geq 30\times$ and ACGT breadth at $10\times \geq 0.90$. The primary CN depth profile excluded duplicate, secondary, supplementary, QC-failed and unmapped alignments; overlapping mates were collapsed for depth. Duplicate-included depth was used only as a sensitivity track.

S16.5 Copy-number screen

Copy-number analysis was restricted to the 13 previously audited genes and used same-contig local-flank normalisation to reduce replication-gradient and contig-level depth bias. The local baseline design used flanks outside a 5-kb inner exclusion and within a 25-kb outer limit; a qualified baseline required at least 20 kb total eligible sequence and at least 5 kb from each side. Candidate classification required technical pass plus all four prespecified 1.50 criteria: ID005 primary local-normalised depth ≥ 1.50 , primary ID005/ID003 ratio ≥ 1.50 , ID005 duplicate-included sensitivity local-normalised depth ≥ 1.50 , and sensitivity ID005/ID003 ratio ≥ 1.50 . Thresholds were not changed after depth values were opened.

S16.6 Targeted minor-allele screen

The targeted universe comprised *gyrA*, *gyrB*, *parC*, *parE*, the 13 coding-audit genes and the four frozen *ompF* noncoding positions, giving 21 objects per isolate and 42 isolate-specific objects in total. A technically eligible position required MAPQ ≥ 30 , base quality ≥ 30 and filtered depth ≥ 100 . A preliminary sequence-level minor-allele signal required at least 10 alternate reads, VAF between 0.10 and 0.40 inclusive, and at least three alternate reads on each strand. No genome-wide exploratory minor-variant scan was authorised. A WGS minor-allele signal would not by itself establish phenotypic heteroresistance.

S16.7 Acquired-quinolone raw-read family resolution

The raw-read acquired-quinolone assay used the frozen 203-sequence/12-family panel. Paired reads were mapped with BWA-MEM -a. Supplementary alignments were excluded and secondary alignments remained eligible. Qualifying alignments required at least 100 alignment columns and identity ≥ 0.95 . Family resolution was based on the unique top family with an alignment-score margin ≥ 5 over the next family. For external *qnrB* contextualisation, a positive raw-read signal required at least 10 resolved fragments, at least three R1 and three R2 supporting ends, breadth ≥ 0.50 and at least five distinct starts. Missing/provenance-unresolved reads or mapping/parser failures were technical missingness, not biological negatives.

S16.8 AAC6Ib variant-class discrimination

The AAC subgate operated only on the frozen ID005 AAC-supporting end universe (31 supporting ends from 20 fragments). Supporting ends were competitively evaluated against frozen CR and non-CR reference classes. Final precedence classified the endpoint as mixed/conflicting if a class-conflict fragment was present or if both CR- and non-CR-informative fragments occurred. Under the observed endpoint, no single aac(6⁺)-Ib-cr or non-CR variant call was supported, and the broad AAC6Ib-family signal was not retained as an acquired quinolone target candidate.

S16.9 Focal qnrB reconstruction and bounded context analysis

The focal qnrB reconstruction was restricted to the 11 phase-B surviving fragments and their 20 original qnrB-supporting read ends; no genome-wide or iterative read recruitment was permitted. The fixed reference was qnrB1 (NG_050469.1; 645 bp). Supporting ends were mapped separately with BWA-MEM -a. Qualifying alignments required ≥ 100 alignment columns and identity ≥ 0.95 . Reference-position base counts required base quality ≥ 30 ; a high-confidence consensus position required depth ≥ 3 and dominant allele fraction ≥ 0.80 . The resulting locus had 12 distinct starts, complete breadth at depths 1 and 2, 0.99845 breadth at depth 3, no high-depth mixed positions and no indel observations. A bounded pair-context subgate evaluated exactly two non-supporting mates and found no unique assembly-context anchor. Consequently, raw-read qnrB locus evidence was retained, but genomic linkage and definitive genomic presence were not established.

S16.10 External ompF contextualisation

The external ompF assay used exactly the six frozen external assemblies and a fixed 601-bp MG1655 NCUNIT05 query spanning positions 986983-987583. Eligible locus alignments required identity ≥ 0.95 and all four fixed candidate positions (query positions 201, 273, 507 and 580) mapped to non-gap assembly bases. The resolved locus was the unique highest-scoring eligible contig/start/strand configuration. Any uncalleable fixed site made the isolate technically unresolved. No new ompF position or locus could be added after execution opened.

S16.11 Reporting and stopping rule

The sequence-only case study was prospectively closed after the coding, noncoding, structural, copy-number, targeted minor-allele, AAC and qnrB context branches reached bounded endpoints. No further candidate discovery, threshold relaxation, post-hoc cohort expansion or cross-system causal ranking is part of the current study. External contextualisation is descriptive only. Technical unavailability/unresolution is always reported separately from biological negatives, and phenotype uncertainty remains an explicit limitation.

Final bounded case-study endpoint

ID005 remains a sequence-only genotype-phenotype discordant ciprofloxacin-resistant case with five retained sequence-level candidate items across two systems: four ompF-region noncoding substitutions and one coherent raw-read qnrB locus with unresolved genomic linkage. Neither retained system recurred in the technically evaluable external CIP-R context. Technical missingness is preserved, and the causal basis of the ciprofloxacin-resistant phenotype remains unresolved.